Calvin UNIVERSITY 1876

# COMPUTER SCIENCE AT CALVIN

*Computer Science Department*



## WHAT IS IN THIS FOLDER

**1** ### Computer Science at Calvin

Information about the degrees in Computer Science, Computer Science Plus and Data Science, and about the cybersecurity certificate. Our BCS is accredited by the Computing Accreditation Commission of ABET.

**2** ### Meet the faculty

The professors who teach in the Computer Science Department, their research, and how to reach out to any of them directly.

**3** ### A map for your four years

A semester-by-semester course plan you can tick off, one sheet for each degree.

#### IF YOU HAVE NEVER WRITTEN CODE

**No previous experience in programming is needed!** CS 108, the first course in the major, starts from the beginning. Math, science, English and some coding in high school leave you well equipped.

## TAKE THE NEXT STEP

### VISIT US

Sit in on a class and see the labs.
calvin.edu/visit

### APPLY

calvin.edu/admissions/apply

### ASK US ANYTHING

Write to any professor on the faculty sheet.
computing.calvin.edu

Scan for the department site, current student projects, and the courses you would take in your first year.

DEPARTMENT OF COMPUTER SCIENCE

North Hall · 3201 Burton Street SE, Grand Rapids, MI 49546 · (616) 526-6000

Calvin University 1876



# COMPUTER SCIENCE









## DEPARTMENT DISTINCTIVES

At the Computer Science Department you can major in three ways: the Bachelor of Computer Science (BCS), Computer Science Plus (BS in CS), or Data Science (BS in DS). There are also minors in computer science, data science and data analytics, and a new certificate in cybersecurity. Students work beside professors who research artificial intelligence and publish on how computing is taught. A liberal arts core means you graduate able to write and communicate in addition to coding, and every professor is a committed Christian who brings their faith to bear on teaching and research.

## BACHELOR OF COMPUTER SCIENCE (BCS)

The BCS is ABET-accredited, meeting the same standards as programs at major research universities. Coursework spans algorithms, artificial intelligence, security, databases, networks, and software engineering. Graduates move into software development, networking, and systems work, or on to graduate study at places like Johns Hopkins, MSU, Michigan Tech, Purdue and the University of Sydney.

## COMPUTER SCIENCE PLUS (BS IN CS)

A full computer science degree paired with a second field, for students who want to bring computing to the subject they care about: business, psychology, languages, or another field entirely. Our Strategic Partners Council regularly reviews the curriculum so it tracks the technologies employers actually use, and students may take paid internships and do research with faculty.

## DATA SCIENCE (BS IN DS)

Calvin was among the first Christian institutions to offer data science. Students combine computer science, statistics, and mathematics with four to five courses in a second discipline such as biology, geography, business, economics, or political science, then finish with a capstone project on real data.

## CYBERSECURITY CERTIFICATE NEW FOR FALL 2026

Eighteen credit hours in cryptography, network defense, and system administration, designed around the NSA's Cyber Defense knowledge units and taught by a professor with three decades in network security. BCS students earn the certificate inside their existing major with no added credit hours. The Bureau of Labor Statistics projects this field will grow 21 percent between 2025 and 2035.



## Derek Schuurman

**PROFESSOR · CHAIR** · NH 296

**Focus:** Faith and technology, embedded systems and IoT, robotics and computer vision

**Email:** derek.schuurman@calvin.edu

- PhD, McMaster University
- BASc and MASc in Electrical Engineering, University of Waterloo
- Writes on the ethics and philosophy of technology



## Ken Arnold

**ASSOCIATE PROFESSOR** · NH 298

**Focus:** Data science, artificial intelligence and Christian faith

**Email:** ken.arnold@calvin.edu

- PhD in Computer Science, Harvard University, 2020
- SM, MIT Media Lab; BS in Electrical and Computer Engineering, Cornell
- Language models for writing, and public understanding of what AI can and cannot do



## Eric Araújo

**ASSOCIATE PROFESSOR** · NH 293

**Focus:** Agent-based modelling, complex networks, computational social science

**Email:** eric.araujo@calvin.edu

- PhD in Computer Science, Vrije Universiteit Amsterdam
- MS, Federal University of Minas Gerais; BS, Federal University of Viçosa, Brazil
- Studies how behaviour, health and misinformation spread



## Rocky Chang

**PROFESSOR** · NH 299A

**Focus:** Network security, internet infrastructure, computational thinking

**Email:** rocky.chang@calvin.edu

- PhD in Computer and Systems Engineering, Rensselaer Polytechnic Institute
- More than 30 years teaching computer science
- Leads the cybersecurity certificate



## Victor Norman

**ASSOCIATE PROFESSOR** · NH 297

**Focus:** Computer science education, K-12 outreach, computing for Christian missions

**Email:** victor.norman@calvin.edu

- PhD and MS, Purdue University; BS, Calvin College
- 14 years in industry as a principal software engineer before teaching
- Works on applying the virtues to how we write programs



## Fernando Santos

**ASSISTANT PROFESSOR** · NH 291

**Focus:** Data science, computational modelling, human-computer interaction

**Email:** fernando.santos@calvin.edu

- PhD in Computational Science, University of São Paulo
- Also works on the philosophy of technology
- Originally from Poços de Caldas, Brazil



## Chris Wieringa

**SYSTEMS ENGINEER** · SB 319

**Focus:** The labs, the servers and the systems every course here runs on

**Email:** chris.wieringa@calvin.edu

- Runs the labs, servers and systems behind every computer science course

### ALSO TEACHING WITH US

**Rob Bobeldyk** Information Technology rob.bobeldyk@calvin.edu
**Stacy DeRuiter** Mathematics and Statistics stacy.deruiter@calvin.edu
**Monica Groenenboom** Engineering monica.groenenboom@calvin.edu
**David Meyer** Adjunct faculty david.meyer@calvin.edu
**Jim Momeyer** Contributing faculty jim.momeyer@calvin.edu
**Brian Paige** Contributing faculty brian.paige@calvin.edu
**Michael Poff** Contributing faculty michael.poff@calvin.edu

Calvin UNIVERSITY 1876

LEADING THE WAY TO A BETTER FUTURE

*computing.calvin.edu*



# COMPUTER SCIENCE DEPARTMENT

## CHRIST-CENTERED UNDERSTANDING

Technology is not neutral. This is part of how we teach you to build new tools for the world. You will be challenged by the many discussions on ethics and philosophy of technology as you progress in your knowledge of computer science. Faith is not an appendix, but the core of what we learn together here.

## PROFESSORS WITH OPEN DOORS

Every course in the major is taught by a professor, and their offices sit along one corridor in North Hall. Their doctorates come from Harvard, Purdue, McMaster, Rensselaer, Amsterdam and São Paulo, and several worked in industry first.

## FOUR-YEAR PATH TO GRADUATION

The Bachelor of Computer Science is 124 credit hours when paired with the Calvin Core, so you graduate with a breadth of knowledge as well as depth in your field. The degree is ABET accredited, the standard employers and graduate schools look for.

ABET | Computing Accreditation Commission

## ACADEMIC OPTIONS

### DEGREES

Bachelor of Computer Science (BCS)
ABET accredited.

BS in Computer Science Plus
Computer science with an applied field of your choice.

BS in Data Science
Computer science and statistics, paired with a subject area.

### MINORS

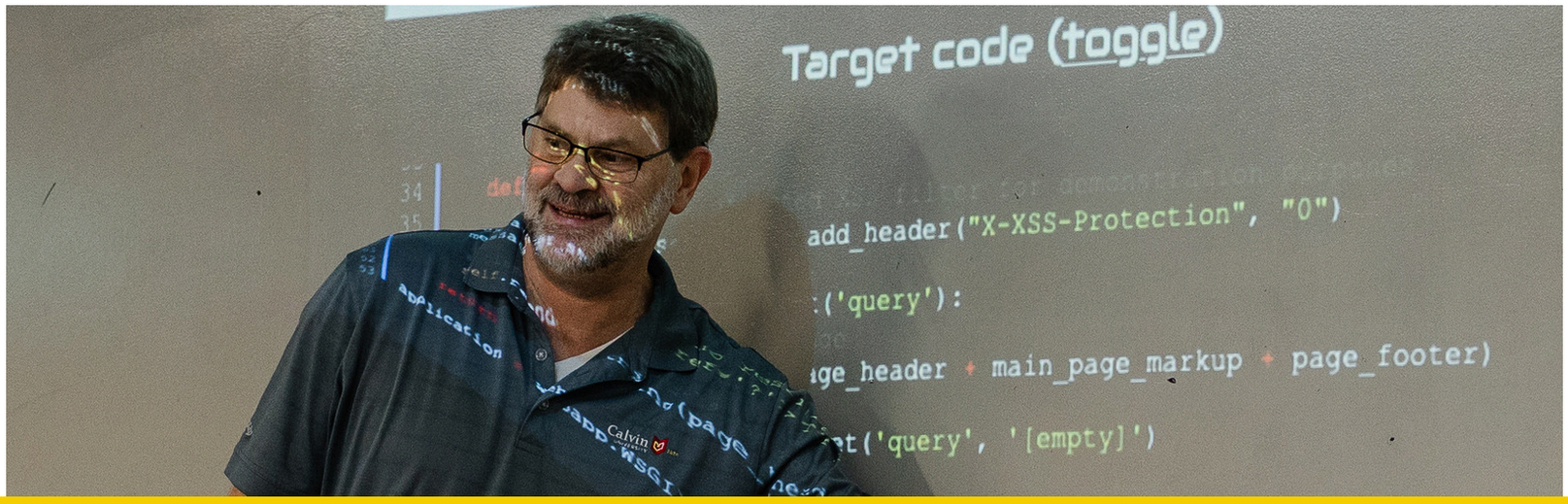
Data Science
Data Analytics

### CERTIFICATE

Cybersecurity
Six courses, 18 credit hours, taken with any of the degrees.

### PAIR WITH

Biology · Business · Economics · Geography · Political science · Psychology · Languages




# WHAT YOU DO HERE

## INTERNSHIPS THAT LEAD SOMEWHERE

Students work a paid internship at a local firm or a national one. For some students, internships become full-time jobs. Recent placements include AMD, Amazon Web Services, General Motors, Corewell Health, Navy Federal Credit Union, Meta, Vervint, Bindplane, EduStaff, Sonset Solutions, Wycliffe Bible Translators, and the Christian Classics Ethereal Library.



## UNDERGRADUATE RESEARCH OPPORTUNITIES

You can join a professor's research already as an undergraduate student. Current research spans language models and everyday creativity, internet infrastructure and security, agent-based models of how misinformation and health behaviour spread, education tools, high performance computing, artificial intelligence, and embedded systems.



## LIBERAL ARTS AND THE CALVIN CORE

Employers want people who can size up a problem, think it through, explain it in writing, and work well with colleagues, clients, and the people they report to. That is what the core builds, and it is woven throughout your undergraduate experience.



## WHERE OUR GRADUATES GO

Alumni are at Google, Amazon, Meta, Steelcase, Corewell Health, Bank of America, the Washington Post, Amway and Pray.com. Others have gone on to graduate school at Johns Hopkins, MSU, Michigan Tech, Purdue, the University of Waterloo, and the University of Sydney, and some now teach at Grand Valley State University and George Mason University. Some have stayed and joined the Christian Classics Ethereal Library and hymnary.org.



# EDUCATION & COURSEWORK

## FOUNDATIONS: YEARS 1 AND 2

You start writing programs in your first semester, in CS 108, and CS 112 takes you into data structures the semester after. The first two years build the craft and the theory underneath it: programming, the machine below the language, and the mathematics that lets you prove an algorithm works rather than hope it does. Core courses run from the first year too, so the questions about what computing is for start early.




## DEPTH: YEARS 3 AND 4

In the upper years you choose the direction: systems and networks, artificial intelligence and data, security, software engineering, or the applied area of your choice. The fourth year goes into one substantial project, the one employers and graduate schools ask about.



### CALVIN HACKATHON

- Students form teams and create a project from scratch in 24 hours
- Students also compete off campus. A Calvin pair took first place at the Atomic Object games

### CYBERSECURITY CERTIFICATE

- Six courses, 18 credit hours, taken with any degree
- Built around the Cyber Defense knowledge units published by the NSA and NCAE-C

### SEMINARS

- People working in the field come in to talk about jobs, careers, and where computer science, data science and AI are heading
- They run through all four years, not only the final one

### SENIOR PROJECT

- A full-year project for real
- Teams choose a project, then plan, research, build, test and deploy it over the year
- Some teams do a research study instead
- Every project ships a public website, and this year a poster too

*The full course list for each degree is on the model program sheets in this folder, one per degree.*

# MORE THAN THE CODE

You write code in your first semester and build systems soon after, always thinking of the impact of your work in the light of a Christian Reformed tradition. Artificial intelligence has not retired that question, it has sharpened it. Faith holds together the craft of building something that works and the wisdom to know how to shape it.

## ADVICE FOR HIGH SCHOOL

Take mathematics as far as your school offers it, and take English and the sciences seriously. This degree asks you to reason, write, and argue as much as to code. If your school teaches programming, that can provide additional preparation.

## WHAT THE DEGREE LEADS TO

- Software engineering
- Artificial intelligence and machine learning
- Cybersecurity and computer forensics
- Data science and analytics
- Cloud architecture and infrastructure
- Databases and information systems
- Embedded systems

## STUDENT ORGANIZATIONS

- Abstraction
- Women in Computing
- Google Developer Student Club

Game nights, Lego events, scavenger hunts and sticky-note pixel art.

## MENTORED FROM THE START

Every student has a faculty mentor from the first semester. You meet through your four years, so there is someone who knows your work when you need advice or a reference.

## CONFERENCES AND COMPETITIONS

Women in Computing takes students to the Grace Hopper Celebration, the largest gathering of women in computing anywhere, and faculty travel with them. Other conferences include:

- Calvin Hackathon
- GrrCon and GR Tech Week
- CCSC and the Create Conference



Calvin University 1876

**MODEL PROGRAM**
*Computer Science Department*

# Bachelor of Computer Science

Degree accredited by the Computing Accreditation Commission of ABET.

## FRESHMAN

| FALL | 14 hrs |
|---|---|
| 4 | **CS 108** Introduction to Computing F,S |
| 4 | **MATH 161** Calculus: Modeling and Computation F,S |
| 0 | **CS 195** Introductory Computing Seminar F,S |
| 2 | **CORE 100** Community and Commitments |
| 4 | **World Languages 1** Waived with two years of high school language at B or better |

| SPRING | 15 hrs |
|---|---|
| 4 | **CS 112** Introduction to Data Structures F,S |
| 2 | **MATH 251** Discrete Mathematics I S |
| 0 | **CS 195** Introductory Computing Seminar F,S |
| 4 | **ENGL 101** Foundational Writing |
| 4 | **World Languages 2** |
| 1 | **Health and Movement 1** |

## SOPHOMORE

| FALL | 16 hrs |
|---|---|
| 4 | **CS 212** Data Structures and Algorithms F |
| 4 | **MATH 255** Introductory Linear Algebra F |
| 4 | **DATA 202** Introduction to Machine Learning F |
| 0 | **CS 195** Introductory Computing Seminar F,S |
| 4 | **Christian Foundations 1** |

| SPRING | 16 hrs |
|---|---|
| 4 | **CS 214** Programming Language Concepts S |
| 2 | **MATH 252** Discrete Mathematics II S |
| 4 | **STAT 243** Statistics S |
| 4 | **COMM 101** Oral Rhetoric |
| 2 | **Christian Foundations 2** |

## JUNIOR

| FALL | 16 hrs |
|---|---|
| 4 | **CS 262** Software Engineering F<br>needs junior standing |
| 4 | **ENGR 220** Introduction to Computer Architecture F |
| 0 | **CS 295** Computing Seminar F,S |
| 4 | **Core course** Knowledge and Understanding |
| 4 | **Elective** |

| SPRING | 15 hrs |
|---|---|
| 4 | **CS 232** Operating Systems and Networking S |
| 4 | **CS elective** From the advanced list |
| 0 | **CS 295** Computing Seminar F,S |
| 1 | **Health and Movement 2** |
| 4 | **Core course** Knowledge and Understanding |
| 2 | **Elective** |

## SENIOR

| FALL | 16 hrs |
|---|---|
| 2 | **CS 396** Senior Project in Computing F |
| 0 | **CS 295** Computing Seminar F,S |
| 2 | **CS elective** |
| 4 | **Science elective** Laboratory science |
| 4 | **Core course** Knowledge and Understanding |
| 4 | **Elective** |

| SPRING | 16 hrs |
|---|---|
| 2 | **CS 398** Senior Project in Computing II S |
| 4 | **CS 384** Perspectives on Computing S |
| 4 | **CS elective** |
| 2 | **Science elective** |
| 4 | **Elective** |

**F** offered in the autumn **S** offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
**124 hours**

# Computer Science Plus

Applied computer science in the field of your choice.

## FRESHMAN

### FALL — 14 hrs

- 4 **CS 108** Introduction to Computing F,S
- 4 **MATH 161** Calculus: Modeling and Computation F,S
- 0 **CS 195** Introductory Computing Seminar F,S
- 2 **CORE 100** Community and Commitments
- 4 **World Languages 1** Waived with two years of high school language at B or better

### SPRING — 15 hrs

- 4 **CS 112** Introduction to Data Structures F,S
- 2 **MATH 251** Discrete Mathematics I S
- 0 **CS 195** Introductory Computing Seminar F,S
- 4 **ENGL 101** Foundational Writing
- 4 **World Languages 2**
- 1 **Health and Movement 1**

## SOPHOMORE

### FALL — 16 hrs

- 4 **CS 212** Data Structures and Algorithms F
- 4 **DATA 202** Introduction to Machine Learning F
- 4 **Your applied field** Twelve hours in total
- 0 **CS 195** Introductory Computing Seminar F,S
- 4 **Christian Foundations 1**

### SPRING — 16 hrs

- 4 **CS 214** Programming Language Concepts S
- 2 **MATH 252** Discrete Mathematics II S
- 4 **STAT 243** Statistics S
- 4 **Core course** Arts and Rhetoric
- 2 **Christian Foundations 2**

## JUNIOR

### FALL — 16 hrs

- 4 **CS 262** Software Engineering F
  needs junior standing
- 4 **ENGR 220** Introduction to Computer Architecture F
- 4 **Your applied field**
- 0 **CS 295** Computing Seminar F,S
- 4 **Core course** Knowledge and Understanding

### SPRING — 15 hrs

- 4 **CS 232** Operating Systems and Networking S
- 4 **CS elective** Six hours in total
- 0 **CS 295** Computing Seminar F,S
- 1 **Health and Movement 2**
- 4 **Core course** Knowledge and Understanding
- 2 **Elective**

## SENIOR

### FALL — 16 hrs

- 2 **CS 396** Senior Project, or CS 383 Practicum F
- 0 **CS 295** Computing Seminar F,S
- 4 **Your applied field**
- 4 **Core course** Knowledge and Understanding
- 4 **Elective**
- 2 **Elective**

### SPRING — 16 hrs

- 2 **CS 398** Senior Project in Computing II S
- 4 **CS 384** Perspectives on Computing S
- 2 **CS elective**
- 4 **Elective**
- 4 **Elective**

**F** offered in the autumn **S** offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
**124 hours**

# Data Science

Computer science, mathematics and statistics together, to equip new professionals of this demanding field of work.

## FRESHMAN

### FALL — 14 hrs

- 4 **CS 108** Introduction to Computing F,S
- 4 **MATH 161** Calculus F,S
- 2 **CORE 100** Community and Commitments
- 4 **World Languages 1** Waived with two years of high school language at B or better

### SPRING — 16 hrs

- 4 **CS 112** Introduction to Data Structures F,S
- 4 **STAT 243** Statistics S
- 4 **ENGL 101** Foundational Writing
- 4 **World Languages 2**

## SOPHOMORE

### FALL — 16 hrs

- 4 **DATA 202** Introduction to Machine Learning F
- 4 **MATH 255** Introductory Linear Algebra F
- 4 **Core course** Knowledge and Understanding
- 4 **Christian Foundations 1**

### SPRING — 15 hrs

- 4 **STAT 341** Computational Bayesian Statistics S
- 4 **Core course** Arts and Rhetoric
- 1 **Health and Movement 1**
- 2 **Christian Foundations 2**
- 4 **Elective**

## JUNIOR

### FALL — 16 hrs

- 4 **STAT 245** Applied Data Analysis F
- 4 **CS elective** CS 212 recommended
- 4 **Business elective** Six hours in total
- 4 **Elective**

### SPRING — 15 hrs

- 2 **CS 375** Machine Learning S
- 2 **CS 376** Advanced Machine Learning S
- 2 **DATA 304** Visualizing Data and Models S
- 2 **DATA 385** Topics in Data Science S
- 1 **Health and Movement 2**
- 4 **Core course** Knowledge and Understanding
- 2 **Elective**

## SENIOR

### FALL — 16 hrs

- 2 **DATA 396** Senior Project in Data Science F
- 2 **DATA 385** Topics in Data Science, second topic F
- 4 **Ethics elective** PHIL 205 or BUS 392
- 4 **Core course** Knowledge and Understanding
- 4 **Elective**

### SPRING — 16 hrs

- 2 **DATA 398** Senior Project in Data Science II S
- 4 **CS 384** Perspectives on Computing S
- 2 **Business elective**
- 4 **Elective**
- 4 **Elective**

**F** offered in the autumn **S** offered in the spring · a full-time semester is about 16 hours
This is one way through the degree. Your adviser will build yours with you.

TO GRADUATE
**124 hours**